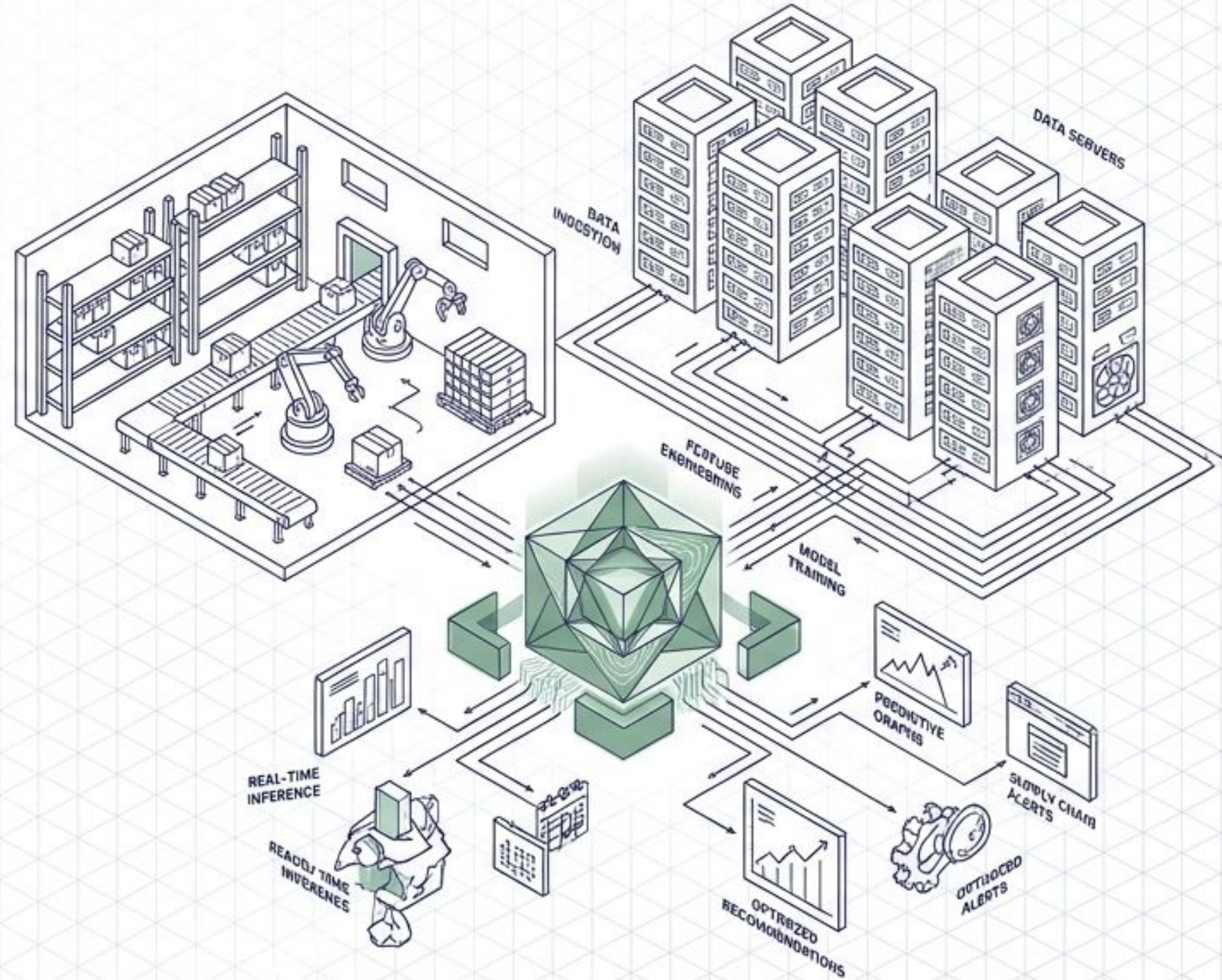


Project Atomic-Link: End-to-End AI Solutions & MLOps

Bloc 4 Defense – RNCP 38777
AI Architect Master

A production-grade supply chain risk prediction engine deployed at Aurora Tech.

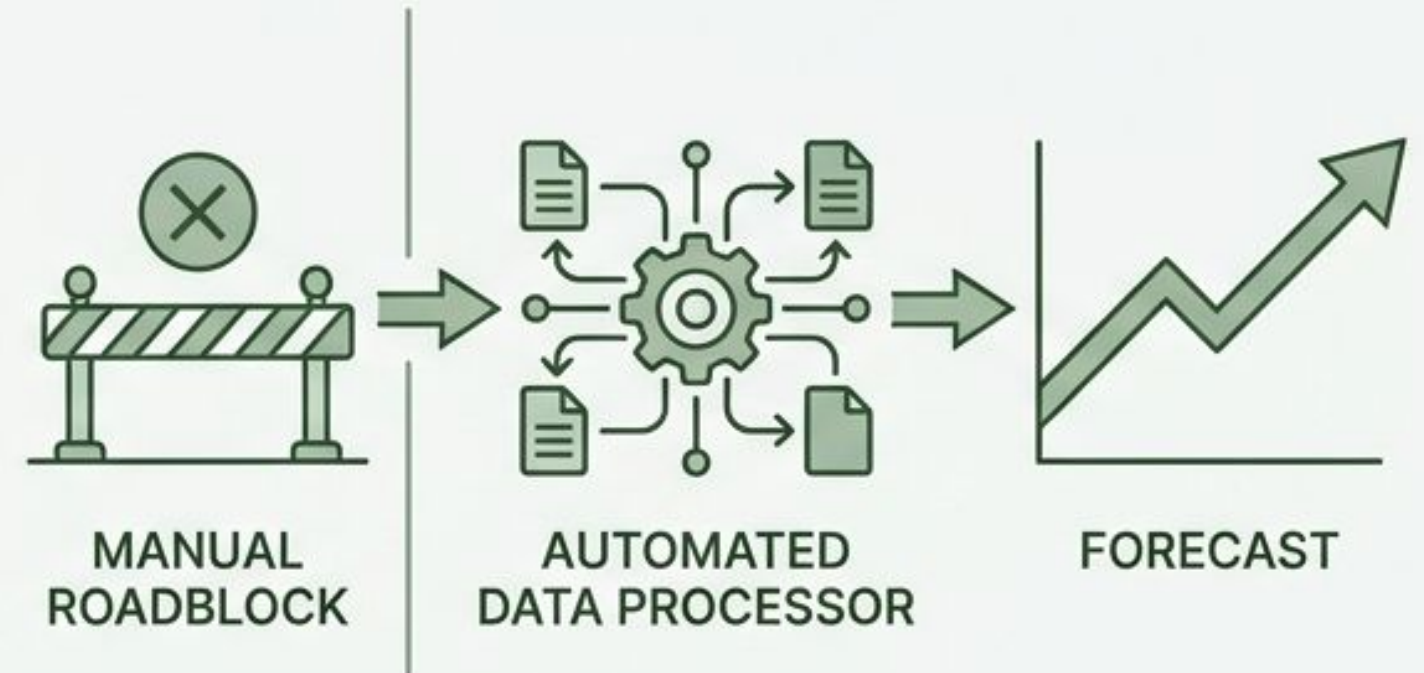


Severe Margin Compression



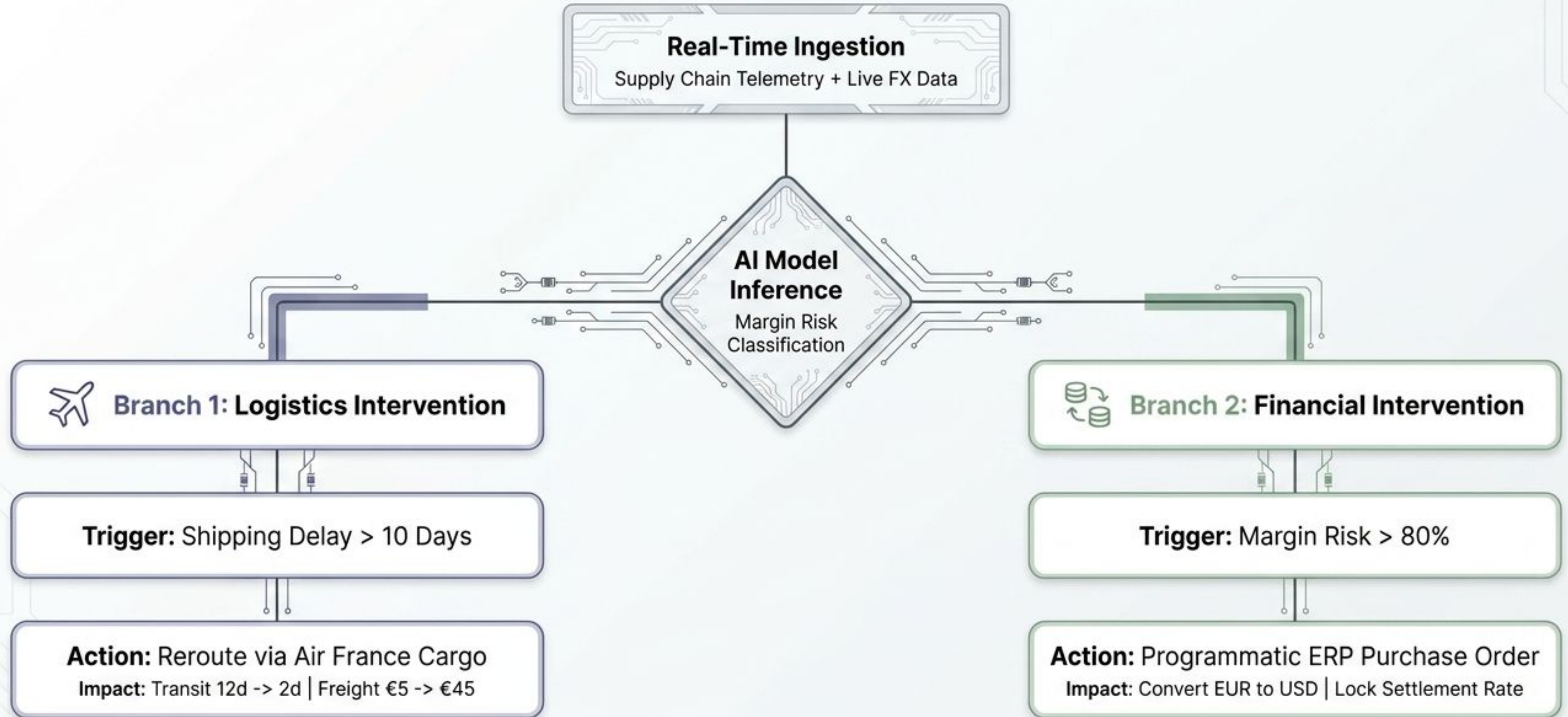
- Global shipping delays bottlenecking critical silicon shipments (e.g., NVIDIA H100 GPUs).
- Extreme macroeconomic currency exposure driven by EUR, USD, and TWD volatility.

Predictive Risk Mitigation



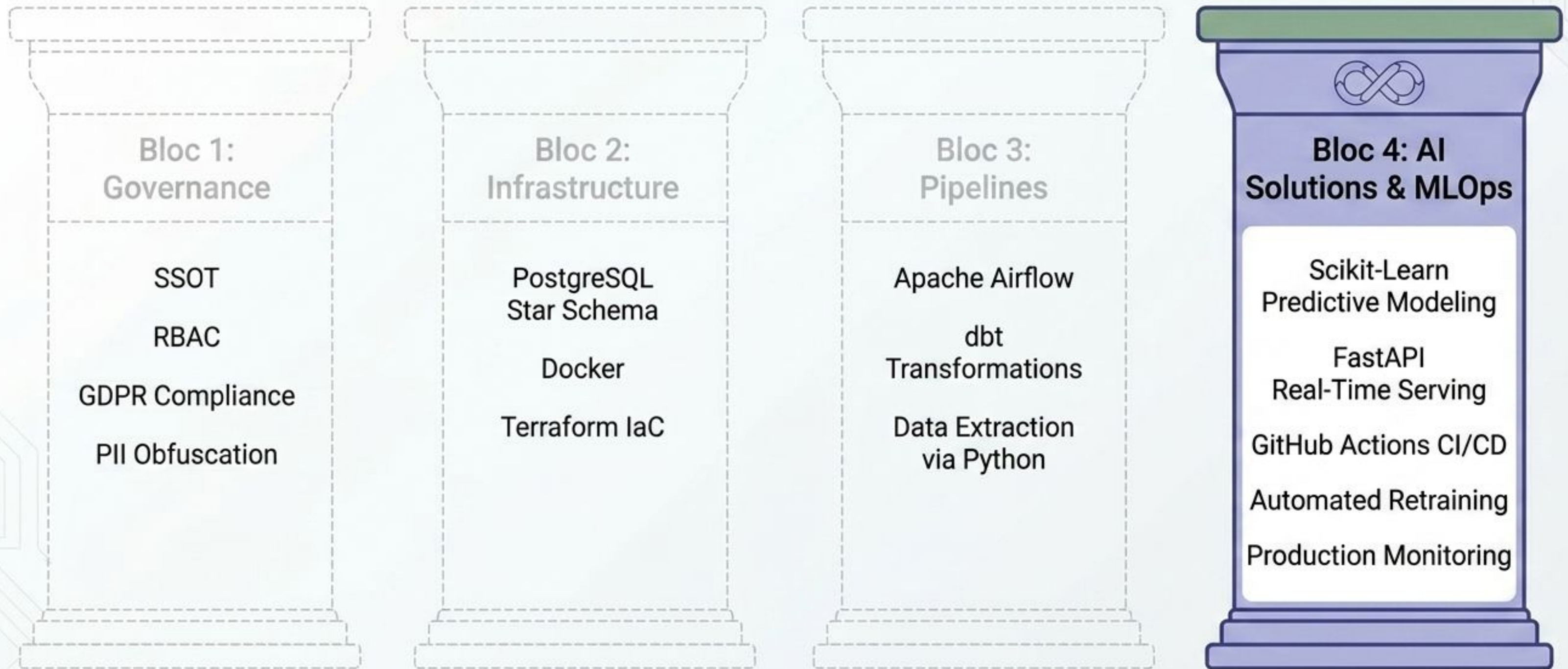
- Predict gross margin degradation before it occurs and programmatically trigger mitigations.
- Minimize logistics OPEX spikes and bypass manual settlement latency.
- Execution speed reduced from 3-5 days to instant automation.

The **Atomic-Link** Intervention Logic

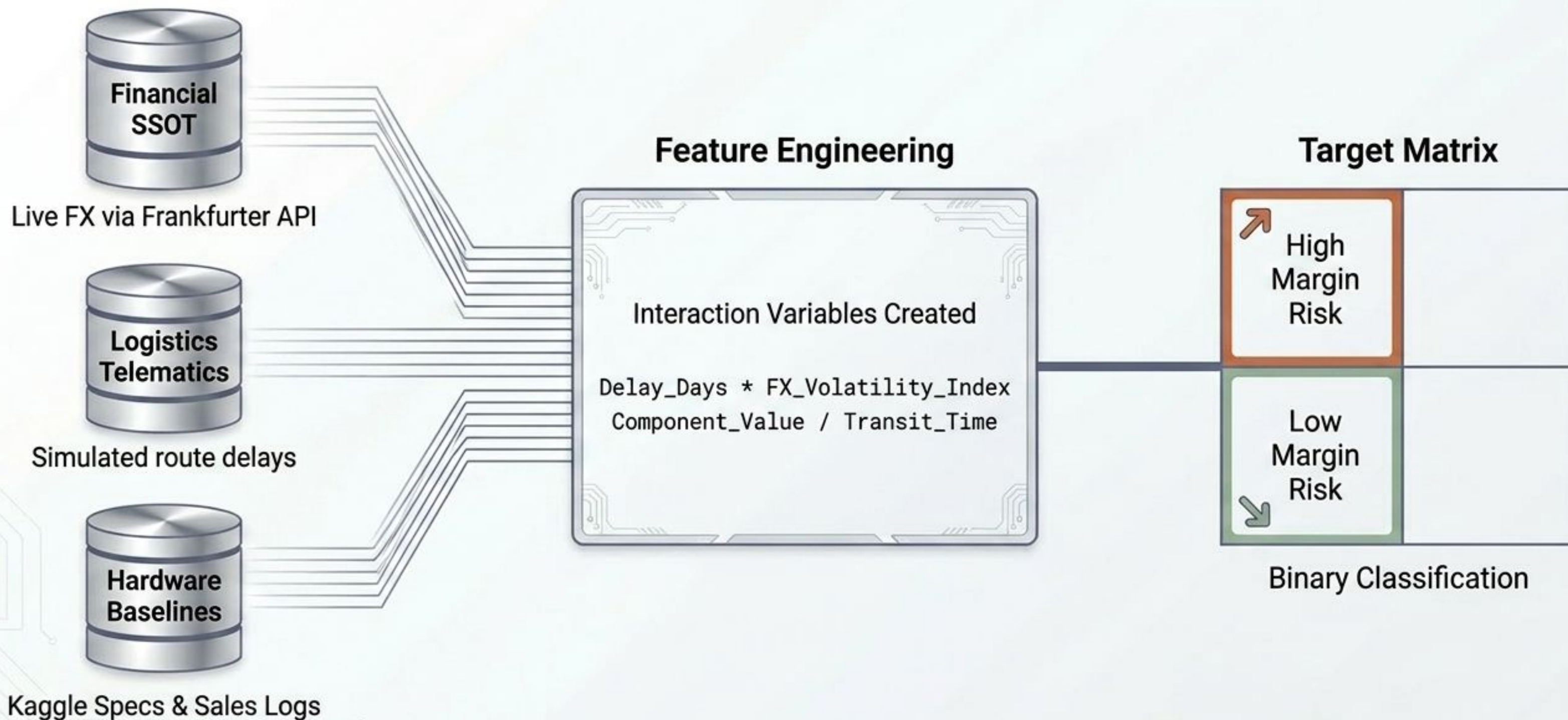


End-to-End Enterprise Architecture

Project Atomic-Link integrates all 4 pillars, with today's defense validating the end-to-end industrialization of Bloc 4.



Data Lineage & Feature Engineering



Algorithm Selection & Training Methodology

Methodology for the Predictive Modeling Component (Bloc 4),
focusing on Explainability and Robustness.

Algorithm Selected

RandomForestClassifier
via Scikit-Learn

Architectural Justification

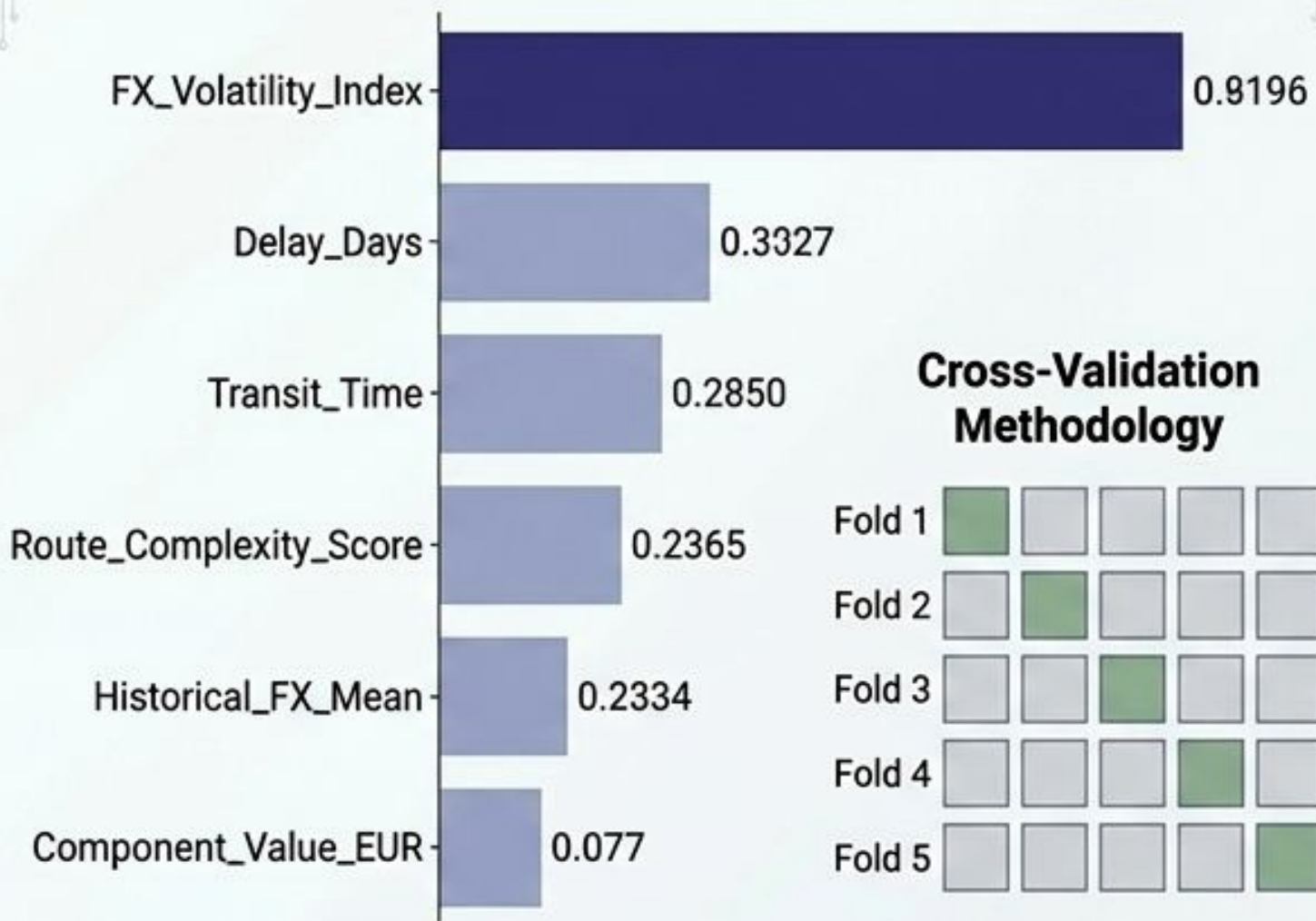


Explainability: Tree-based feature importance is crucial for business stakeholders to trust automated ERP hedging triggers.

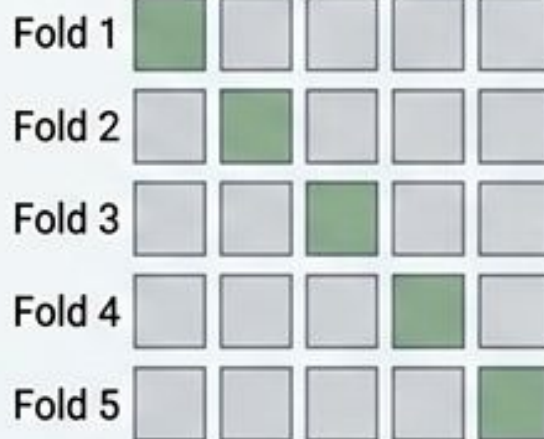


Data Suitability: Robust handling of non-linear, heterogeneous tabular data combining shipping route metrics with currency indices.

Feature Importance



Cross-Validation Methodology



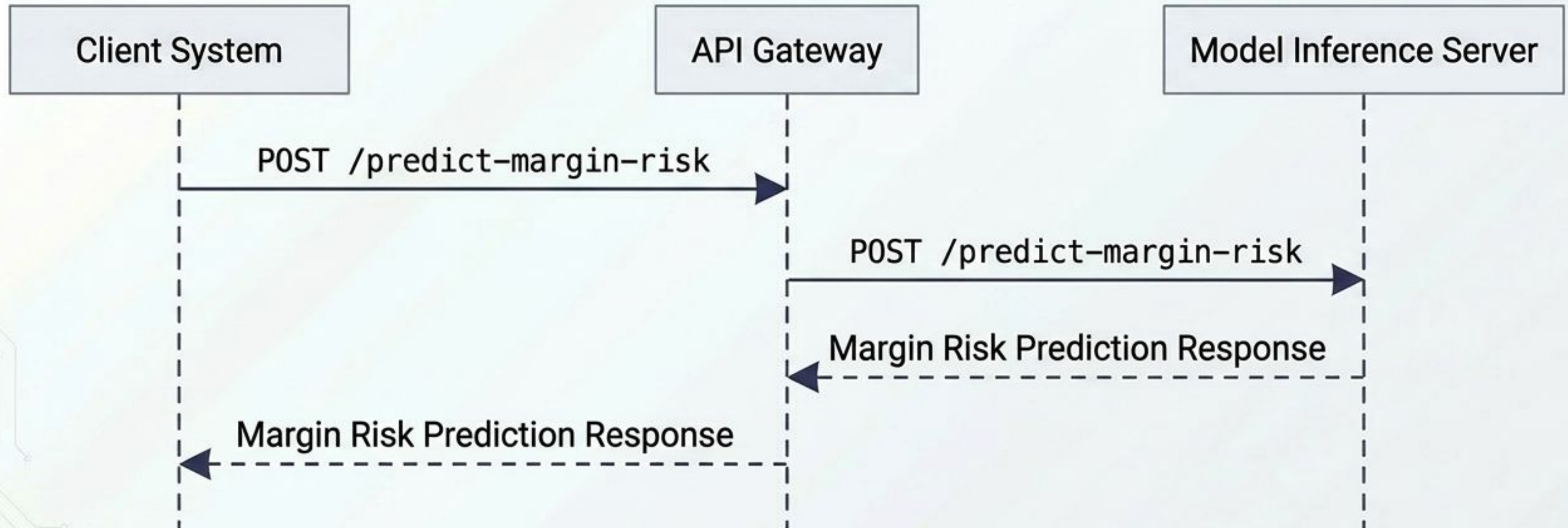
Training Architecture

- Modularized codebase using /scripts and /src directories.
- Rigorous cross-validation methodology employed to prevent overfitting on historical FX spikes.

Serving API Architecture

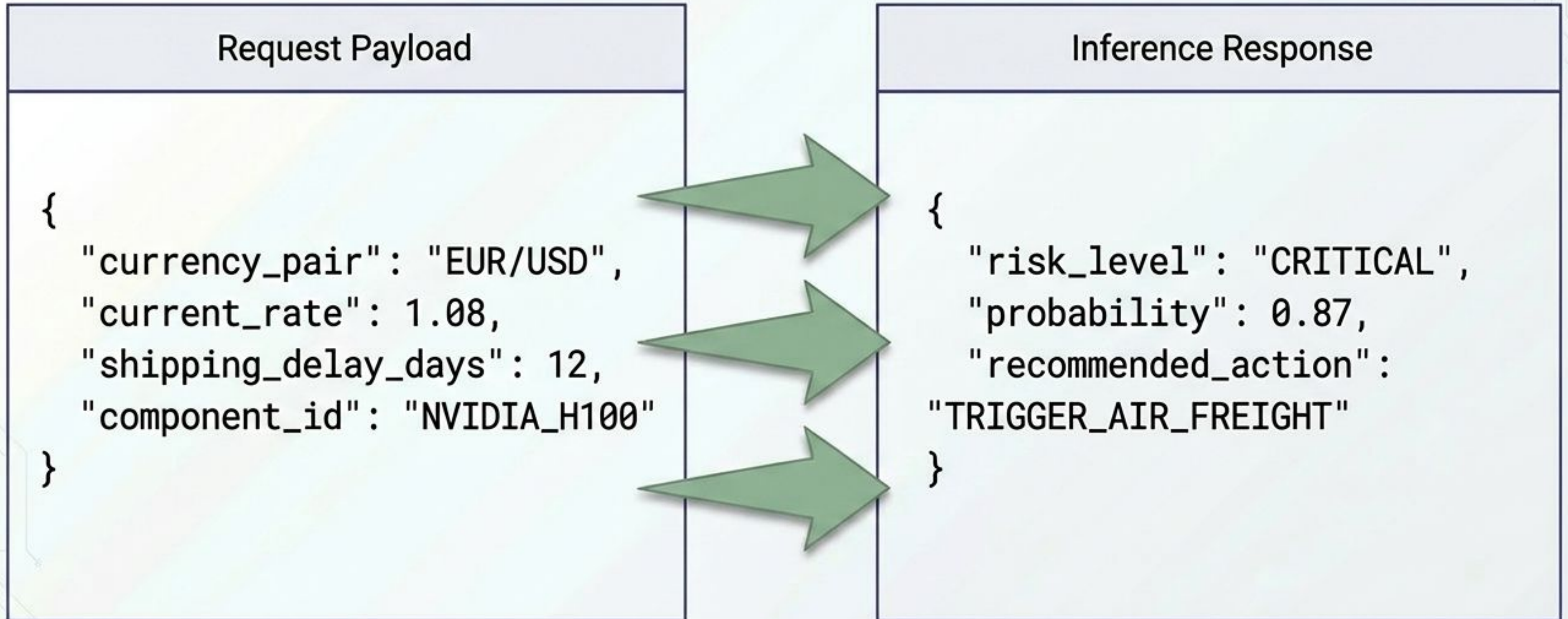
FastAPI framework chosen for asynchronous capabilities and high-performance native Python integration.

Fully Dockerized environment guarantees strict staging-to-production parity.



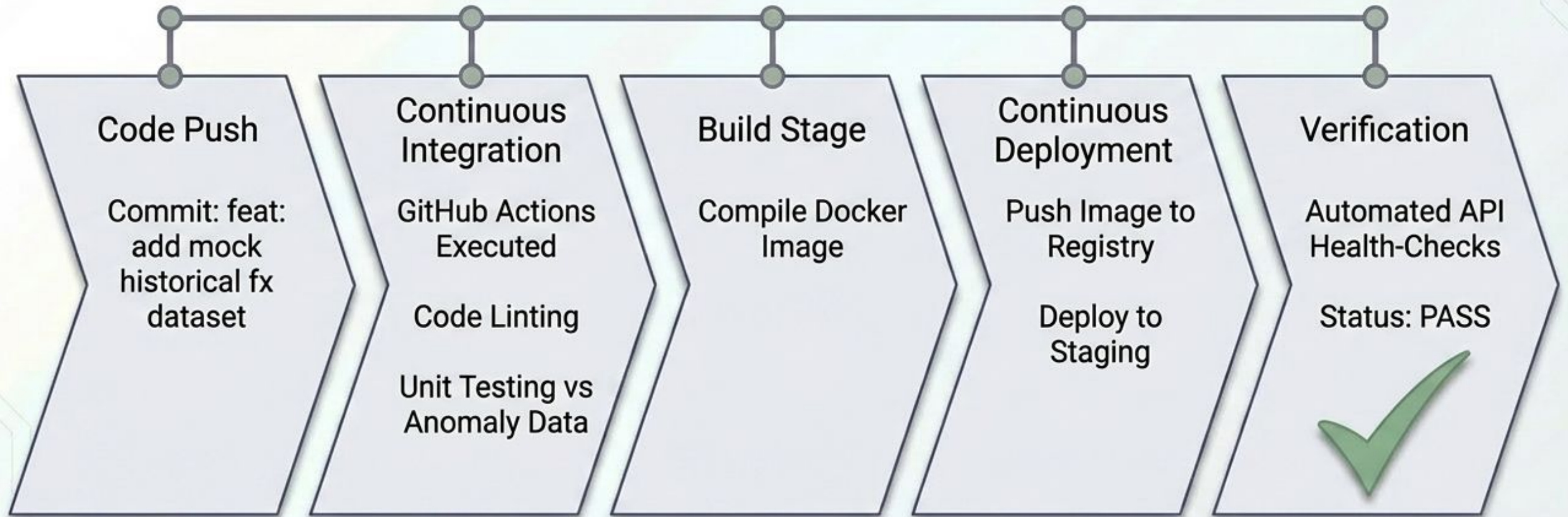
Payload & Response Structure

The API translates mathematical probability directly into actionable ERP and routing commands.



Automated Deployment Flow

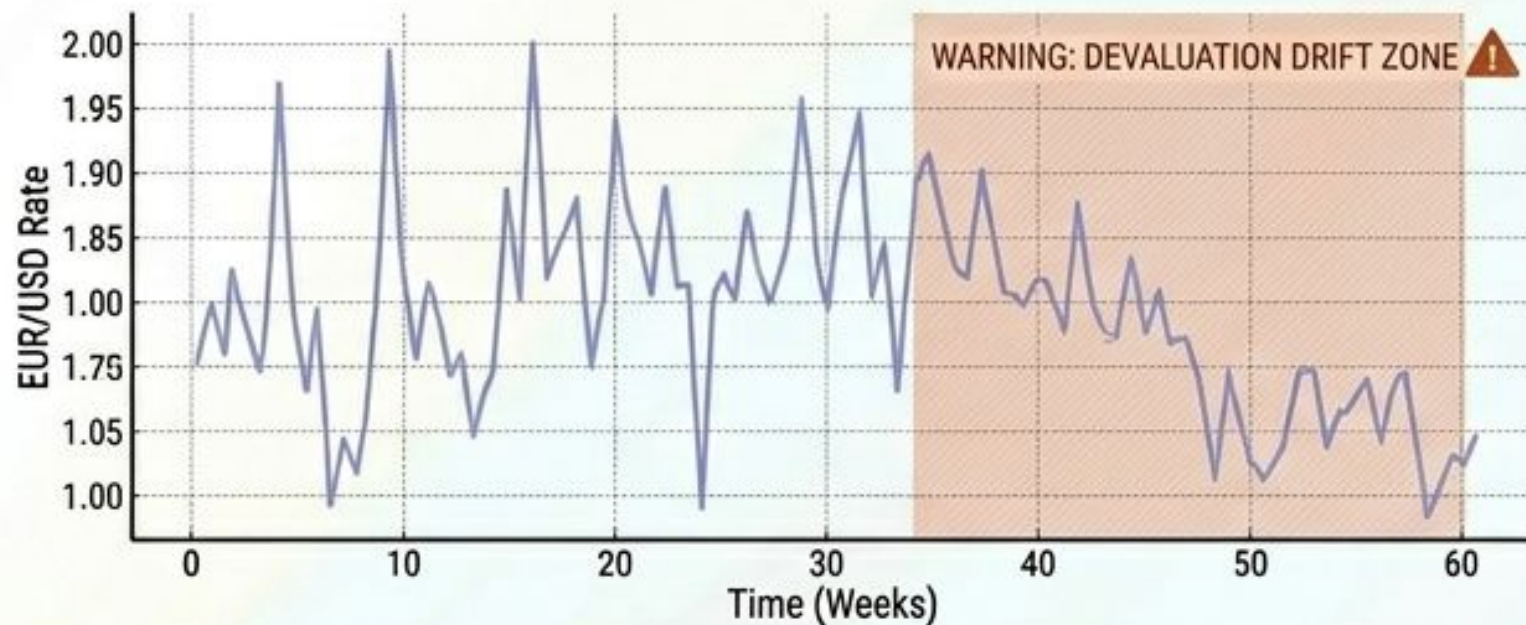
Rigorous left-to-right linear pipeline.



Production Monitoring: Drift & Health

Data Drift Detection

Continuous tracking of massive macro-economic shifts, identifying sudden Euro devaluation against the USD.



Concept Drift Monitoring

Evaluates whether the real-world relationship between shipping delays and freight costs fundamentally shifts.

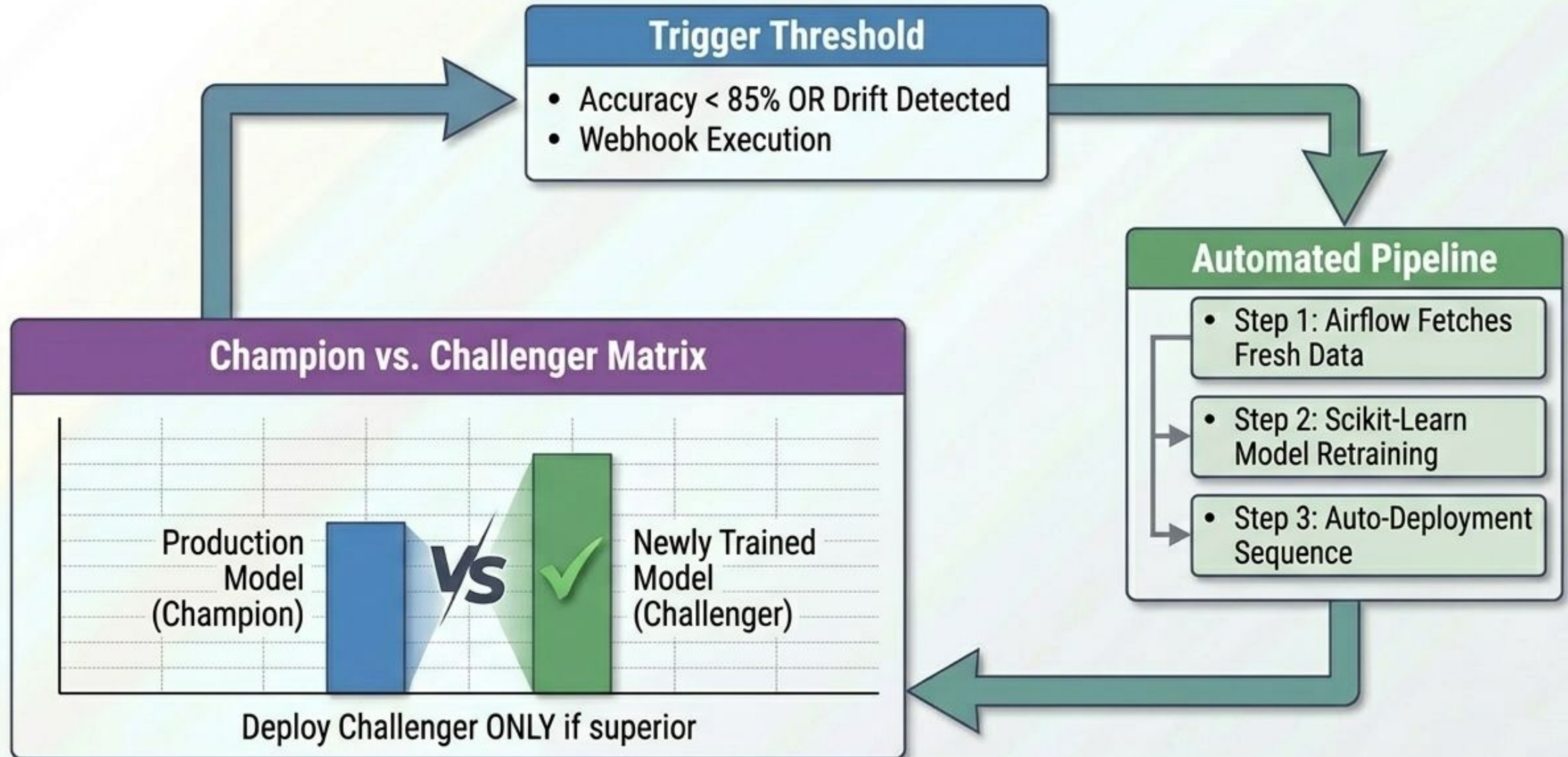


Infrastructure Health Metrics

Real-time tracking of API endpoint latency and HTTP 400/500 error rates to ensure high availability for ERP integration.



The Self-Healing System: Automated Retraining

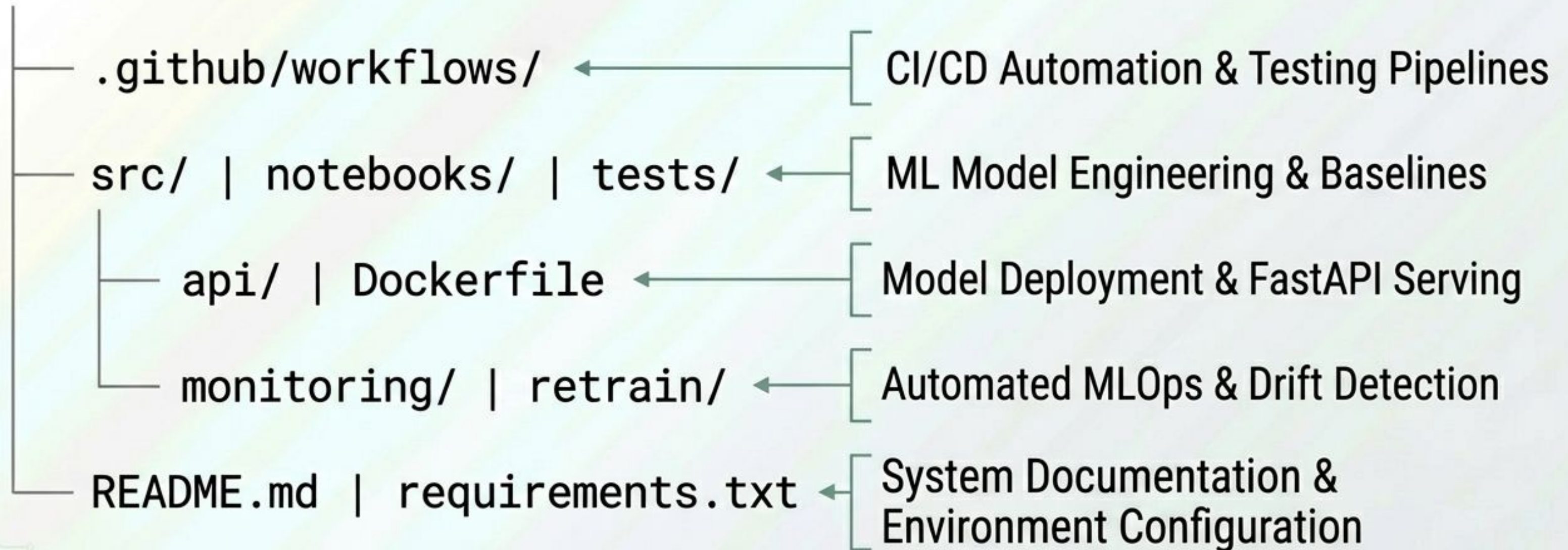


System Diagnostic: Traditional vs. Atomic-Link

Operational Feature	Traditional SCM	Atomic-Link AI Platform
Data Integration	Siloed (Logistics separate from Finance)	Unified (Bloc 2 Data Warehouse)
Margin Protection	Reactive (Post-mortem analysis)	Predictive (Machine Learning foresight)
Intervention Speed	Manual ERP Settlement (3-5 business days)	Automated API Hooks (Instant execution)
System Maintenance	Manual data dumps & retraining	Automated CI/CD & continuous drift detection

Project Repository & MLOps Organization

Repository structured to strict enterprise deployment standards, ensuring seamless hand-off and scalability.



Final Synthesis & Business Impact

Technical Achievement

- ✓ Successfully industrialized an end-to-end AI solution—from feature engineering and FastAPI serving to rigorous CI/CD and automated model retraining.



Business Impact

- ✓ Protected critical hardware profit margins from FX and logistics volatility.
- ✓ Modernized Aurora Tech's IT infrastructure.
- ✓ Established a resilient, scalable MLOps culture.

INPUT FEATURE VECTORS

JSON PAYLOAD

Currency Pair (EUR/USD)1.080

WEAK RANGEPARITYSTRONG

Transit Delay Expected+2 Days

ON TIMESEVERELY DELAYED

Freight Class Override

Standard Ocean

Expedited Air

 RUN PREDICTION

 MARGIN AT-RISK PROBABILITY

5.0%

 **SYSTEM NOMINAL**

CONFIDENCE: 94.2%

Vectors aligned with historical positive-margin executions. No strategic intervention recommended.

TOP DRIVING FEATURES (SHAP VALUES)

Freight Class		-0.08
EUR/USD Pair		+0.11